

Université Nice Côte d'Azur

M1 AI
Année 2023/24

Hate speech detection on online network

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19 octobre 2026

Acknowledgements

First and foremost, I extend my deepest thanks to Ocampo Benjamin for his exceptional guidance and unwavering support throughout this journey. His commitment to excellence and his willingness to invest time and effort have greatly enriched my experience. His consistent presence, patient mentoring, and insightful feedback have been pivotal in my development.
Also, I am profoundly grateful to Cabrio Elena for her expert advice and strategic direction. Her profound knowledge and incisive perspectives have significantly shaped my approach to this work. Her ability to provide clear, actionable guidance while encouraging independent thought and analysis has been especially beneficial.

1 Abstract

In the ever-evolving landscape of online communication, an integral component of our daily lives, the prevalence of hate speech poses a significant challenge to the principles of respect and inclusively within digital interactions. This study explores the intricate dynamics of hate speech, employing machine learning techniques, particularly in Natural Language Processing (NLP), to discern explicit, implicit, and nuanced forms.
While explicit hate speech is identifiable through overtly offensive language, the effectiveness of current machine learning methods diminishes when confronted with the subtleties inherent in implicit and nuanced expressions. The study delves into experiments, utilizing the ISHate benchmark,to uncover patterns, assess model behavior, and explore augmentation strategies such as oversampling and perturbation.

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2 Introduction

In the contemporary era, the use of online communication has become an integral aspect of our daily lives, shaping the way we connect, share information, and engage in digital conversations. However, the proliferation of hate speech in online spaces has emerged as a formidable threat, challenging the fundamental principles of respect and inclusively that should characterize digital interactions. Cultivating and preserving healthy online environments becomes paramount, as these spaces wield significant influence in shaping our perceptions, beliefs, and collective social discourse.

In combating hate speech, machine learning techniques, particularly those employed in Natural Language Processing (NLP), have showcased promising capabilities, especially in identifying explicit forms of hate speech. Explicit hate speech, marked by its overtly offensive language, is relatively discernible, exemplified by unequivocal statements such as "Africans are so dumb." The identification of such explicit instances is an essential step towards creating a safer digital landscape.

However, the efficacy of current machine learning methods encounters limitations when confronted with the subtleties inherent in implicit and nuanced forms of hate speech. Distinguishing implicit hate speech poses a more intricate challenge, requiring understanding figurative language elements like irony or sarcasm. For instance, the statement "Are you sure that Islam is a peaceful religion ?" ,the statement "Are you sure that Islam is a peaceful religion ?" is an example of implicit hate speech because it conveys a potentially offensive or derogatory message indirectly. The irony lies in the way the question is framed, creating a subtle insinuation rather than making a direct, overtly offensive statement.

Moreover, the landscape of hate speech expands to include subtle variations, where elusive messages rely on indirect methods to convey meaning. An example such as "As a brit my knowledge of american law is somewhat lacking but even i know that this holder groid has committed treason ", this example is explicit as it literally conveys hateful meanings with the expression holder groid, but it is subtle as the intended hateful meaning is disguised by the first and second underlined constructions, pushing out the attention from the explicit hateful expression.

Our first mission involved a meticulous examination of Hate Speech Target Distribution, aiming to uncover prevalent trends and variations in the labeling of hate speech across various contexts. Subsequently, we delved into the Label Distribution in Relation to Augmentation Methods, aiming to assess how different augmentation strategies impact the distribution and classification of hate speech labels. Our research also sought to evaluate Model Performance by Message Type and Target Group, aiming to discern how machine learning models perform in categorizing hate speech messages based on the nature of the content and the specific target groups involved. Lastly, we explored the Effectiveness of Different Augmentation Strategies, aiming to determine the efficacy of various augmentation techniques in enhancing hate speech detection capabilities.

NOTE : *This report contains examples of language that may be offensive to some readers. They do not represent the views of the authors.*

3 Related Work

Expanding on what we've learned from earlier studies, like the ISHate benchmark by Ocampo and team in 2023 [2], this project takes a closer look at the hidden aspects of hatred. We aim to gain a valuable understanding of the subtle ways hate is expressed online explained in research by El Sherif and others in 2021 [4]. To do this, we're using advanced Natural Language Processing (NLP) techniques, such as BERT and HateBert transformer models.
Imagine NLP as a kind of technology that helps computers understand and work with human language. In our case, we're using it to uncover the emotional side of messages and tell the difference between messages that are explicit and those that are implicit by Tommaso Caselli et al.,2020 [3] . The goal is to move from just talking about ideas to putting those ideas into action. This shift helps us get a more detailed understanding of how hate speech works in different online conversations.
In simpler terms, we're not just thinking about hate speech ; we're actively doing things to better understand it. We're using advanced technology to analyze how emotions are expressed, and we're figuring out the difference between when hate is obvious and when it's not so clear. This move from talking about ideas to taking action helps us see hate speech in a more detailed and complete way.

4 Methodology

4.1 Research Question : Analysis of Hate Speech Target Distribution

In this investigation, our primary objective was to determine the locations where hate speech is most prevalent. We conducted experiments using the ISHate data set, which includes distinct sections for training, development, and testing. To facilitate analysis, we introduced a special category column, identifying the target of the message if it is a women or Muslims or Jews ....
To analyze hate speech targets in the dataset, we employed word embedding and NLP techniques. Then, the Spacy model aided in sentence tokenization and subject (noun) extraction. Furthermore, we generated synonyms for these subjects using the Glove model, storing them as keys in a dictionary. Then, during dataset processing, subjects in the text were replaced by their common values from the dictionary.

4.2 Research Question : Label Distribution in Relation to Augmentation Methods

Understand the distribution of different target groups across augmentation methods. Look for patterns or variations that may suggest the effectiveness of certain methods for specific target groups.

4.3 Research Question : Evaluating Model Performance by Message Type and Target Group

We defined a list of augmentation methods. For each method, a pre-trained model was loaded using the Hugging Face pipeline. Then, predictions were made on the test data, and classification metrics (precision, recall, and F1-score) were calculated for each target group using the

4.4 Research Question : Effectiveness of Different Augmentation Strategies

Next,our focus shifted to understanding how different methods or tricks could alter hate speech labels. Various methods were explored. Then, their impact on label distribution, especially in the "target" column, was analyzed. The objective was to assess the prevalence of different hate speech targets resulting from each method.
Two primary methods were employed. First, balancing the number of implicit and explicit examples by oversampling implicit instances using `resample`. Then, creating additional examples by pairing random rows of implicit speech. Also, the text of these pairs was concatenated to form a a new implicit hate speech examples. After that, our aim was to evaluate how well computers could comprehend different types of hate speech.
Finally, implementing a data augmentation function to generate new instances by combining pairs of texts for the same target class. This method is used to create new training instances by concatenating consecutive text samples. This technique is

commonly used for oversampling to address class imbalance or to introduce diversity in the training data

5 Results

5.1 Analysis of Hate Speech Target Distribution

Label	ASIAN	DISEASE	JEWS	MUSLIMS	WOMEN
HS	294	49	834	735	628
Non-HS	255	17	198	150	729
Explicit HS	246	45	741	681	568
Implicit HS	48	4	93	54	60
Subtle HS	15	3	17	9	20
Non-Subtle HS	279	46	817	726	608

TABLE 1 – Train data distribution

Label	ASIAN	DISEASE	JEWS	MUSLIMS	WOMEN
HS	56	17	191	167	134
Non-HS	65	5	37	29	156
Explicit HS	44	17	172	152	122
Implicit HS	12	0	19	15	12
Subtle HS	54	16	186	165	130
Non-Subtle HS	2	1	5	2	4

TABLE 2 – Test data distribution

The test results reveal a nuanced picture of the model's performance across various target labels. In analyzing hate speech directed towards different groups, such as ASIAN, DISEASE, JEWS, MUSLIMS, and WOMEN, the model exhibits varying levels of accuracy. Notably, it demonstrates effectiveness in identifying explicit hate speech, as evidenced by the precision in labels like JEWS and MUSLIMS. However, challenges arise in detecting implicit hate speech, where the model's performance is less pronounced. These findings highlight the importance of refining the model's capabilities, particularly in discerning subtle and implicit forms of hate speech, to enhance its overall effectiveness in mitigating online toxicity.

5.2 Label Distribution in Relation to Augmentation Methods

Aug method	ASIAN	DISEASE	JEWS	MUSLIMS	WOMEN
AAV	304	8	704	528	592
ALL	1271	75	2376	1761	1757
BT	43	2	31	32	49
EDA	372	29	684	422	470
GM	36	0	42	23	26
GM-revised	16	0	12	7	9
RA	38	7	48	41	35
RI	54	5	141	48	51
RNE	64	0	146	172	53
RSA	344	24	568	488	472

Table 3 – Data distribution for implicite hate speech message

Augm method	ASIAN	DISEASE	JEWS	MUSLIMS	WOMEN
AAV	344	0	344	160	288
ALL	933	97	1000	682	1010
BT	15	0	16	4	20
EDA	277	44	261	149	300
GM	33	0	38	26	21
GM-revised	40	0	41	23	22
RA	11	3	21	21	24
RI	14	2	27	10	35
RNE	41	0	60	65	20
RSA	208	48	192	224	280

Table 4 – Data distribution for subtle hate speech message

The two tables presents the data distribution for implicit and subtle hate speech messages across various augmentation methods, shedding light on the model's performance in identifying subtle forms of toxic content. The augmentation methods, including AAV, ALL, BT, EDA, GM, GM-revised, RA, RI, RNE, and RSA, are evaluated for their impact on recognizing hate speech directed towards ASIAN, DISEASE, JEWS, MUSLIMS, and WOMEN. The results indicate diverse patterns in the distribution of implicit hate speech messages, underscoring the need for careful consideration of augmentation strategies to effectively address this nuanced aspect of online toxicity. These findings contribute valuable insights for refining the model's training approach and augmenting data-sets to enhance its ability to discern implicit forms of hate speech across different target groups

5.3 Evaluating Model Performance by Message Type and Target Group

Labels	ASIAN			JEWS			MUSLIMS	
	P	R	F1	P	R	F1	P	R
Non-HS	0.843750	0.830769	0.837209	0.589744	0.621622	0.605263	0.593750	0.655172
Explicit HS	0.770833	0.840909	0.804348	0.902299	0.912791	0.907514	0.903846	0.927632
Implicit HS	0.666667	0.500000	0.571429	0.666667	0.526316	0.588235	0.625000	0.333333
Non-HS	0.848485	0.861538	0.854962	0.600000	0.648649	0.623377	0.730769	0.655172
Explicit HS	0.755102	0.840909	0.795699	0.900000	0.889535	0.894737	0.900000	0.947368
Implicit HS	0.833333	0.416667	0.555556	0.500000	0.473684	0.486486	0.400000	0.266667
Non-HS	0.850746	0.876923	0.863636	0.578947	0.594595	0.586667	0.666667	0.758621
Explicit HS	0.770833	0.840909	0.804348	0.906433	0.901163	0.903790	0.915033	0.921053
Implicit HS	1.000000	0.500000	0.666667	0.578947	0.578947	0.578947	0.400000	0.266667
Non-HS	0.861538	0.861538	0.861538	0.575000	0.621622	0.597403	0.678571	0.655172
Explicit HS	0.782609	0.818182	0.800000	0.915663	0.883721	0.899408	0.910256	0.934211
Implicit HS	0.900000	0.750000	0.818182	0.636364	0.736842	0.682927	0.583333	0.466667
Non-HS	0.877193	0.769231	0.819672	0.600000	0.648649	0.623377	0.615385	0.551724
Explicit HS	0.775510	0.863636	0.817204	0.914634	0.872093	0.892857	0.901316	0.901316
Implicit HS	0.466667	0.583333	0.518519	0.500000	0.631579	0.558140	0.388889	0.466667
Non-HS	0.838235	0.876923	0.857143	0.568182	0.675676	0.617284	0.600000	0.724138
Explicit HS	0.760870	0.795455	0.777778	0.892216	0.866279	0.879056	0.904459	0.934211
Implicit HS	0.857143	0.500000	0.631579	0.588235	0.526316	0.555556	1.000000	0.266667
Non-HS	0.857143	0.830769	0.843750	0.676471	...			

Table 5 – Hate speech classification per target

The evaluation metrics for precision, recall, and F1 score were computed for the labels ASIAN, JEWS, MUSLIMS and WOMEN across various target groups, revealing the model's performance in distinguishing hate speech from non-hateful content. For the Non-HS (Non-Hate Speech) category, the precision, recall, and F1 score for ASIAN were 0.857143, 0.830769, and 0.843750, respectively. Similarly, the Explicit HS (Explicit Hate Speech) category demonstrated precision, recall, and F1 score values of 0.760870, 0.795455, and 0.777778 for ASIAN. The Implicit HS (Implicit Hate Speech) category presented noteworthy variability, with ASIAN precision, recall, and F1 score standing at 0.857143, 0.500000, and 0.631579, respectively. These outcomes underscore the challenges in accurately classifying implicit hate speech and highlight the need for further refinement in the model's training and evaluation strategies, particularly concerning nuanced forms of toxic content directed at diverse target groups.

5.4 Augmentation strategies

```
data_to_aug = df[df["implicit_layer"] == "Implicit HS"].copy()
augmented_data = augment_data(data_to_aug)
augmented_data
```

	target	text
0	MUSLIMS	the west has let islam take over our countries...
1	MUSLIMS	muslims in britain vote for the labour party ....
2	JEWS	jews secretly run our government . all the jew...
3	WOMEN	feminism is the ideology which suggests that m...
4	WHITE PEOPLE	this is aryan no orthodox christian white bl...
...	...	...
24818	NOT GIVEN	time and again , it has been proven that jews ...
24819	NOT GIVEN	they have nothing to do with our society . tak...
24820	NOT GIVEN	see # hypocrisy ? see our last tweet ! jews ke...
24821	NOT GIVEN	white people do not care people like you fee...
24822	NOT GIVEN	how did women hurt men s bottom and hide their...

24823 rows × 2 columns

FIGURE 1 – Snippet of the new data generated after augmentation method

```
[ ] augmented_data["text"].iloc[2]

'jews secretly run our government . all the jews in most countries are rich and control everything . it is all down to greed and jealousy . they like to co
```

FIGURE 2 – Example of the new text generated

Implementing a data augmentation function to generate new instances by combining pairs of texts for the same target class. This method is used to create new training instances by concatenating consecutive text samples

5.5 General Interpretation

The comprehensive analysis of the model's performance across various target groups and augmentation methods reveals both strengths and areas for improvement. In general, the model exhibits robust performance in identifying explicit hate speech, as indicated by high precision, recall, and F1 scores. This suggests a successful recognition of overtly harmful content targeting different demographic groups. However, challenges arise when dealing with implicit hate speech, where the model's performance shows greater variability.

The results underscore the complexity of distinguishing implicit forms of hate speech, marked by lower precision and recall values. This emphasizes the need for a more nuanced approach and further refinement in addressing the intricacies of implicit toxic content. Augmentation methods also play a role in influencing the model's performance, with variations observed across different techniques.
Additionally, the model demonstrates competence in handling Non-Hate Speech instances, achieving high precision and recall for most target groups. This implies a reliable capability to differentiate between harmful and non-harmful content.

6 Future Work

In the upcoming stages of our research, We aim for incorporating information from the SBIC dataset. This involves a comprehensive analysis of the SBIC dataset to identify valuable additions that align with the objectives of ISHate. We aim to explore various aspects of SBIC, ensuring its compatibility with ISHate. Following this, we plan to identify specific features within the SBIC dataset that can significantly contribute to the improvement of ISHate. Subsequently, we will conduct experiments to seamlessly integrate the selected data from SBIC into the ISHate dataset. The final step involves evaluating the performance of the enhanced ISHate dataset using appropriate metrics, ensuring the effectiveness of the proposed enhancements.

7 Conclusion

In conclusion, the study has provided a comprehensive examination of machine learning models in the context of hate speech detection, particularly focusing on explicit and implicit forms targeting various demographic groups. The findings highlight the model's robustness in identifying explicit hate speech, showcasing commendable precision, recall, and F1 scores. However, challenges persist when dealing with implicit hate speech, necessitating further refinement and attention to the nuanced aspects of such content.

The evaluation of different augmentation methods reveals their impact on model performance, emphasizing the importance of carefully selecting and fine-tuning these techniques. The model's proficiency in discerning Non-Hate Speech instances indicates a strong ability to differentiate between harmful and non-harmful content.

Looking ahead, addressing the complexities of implicit hate speech remains a key area for improvement. Continued research endeavors, coupled with ongoing model optimization and exploration of diverse augmentation strategies, will contribute to advancing the field. The commitment to fostering safer online environments stands paramount, emphasizing the need for continuous efforts in enhancing hate speech detection capabilities and ensuring inclusivity for users across diverse backgrounds.

Overall, this study serves as a valuable contribution to the ongoing pursuit of leveraging machine learning for creating more secure and respectful online spaces.

Références

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